RABBIT PVT. LTD.

MINOR PROJECT ASSESSMENT

Network Discovery, Information Gathering, Enumeration and FTP Security Testing

Assessment Type Scenario-Based Practical Maximum Marks 50

Student Name _____ Date _____

Batch / Team _____ Target Student-selected authorized lab target

SCENARIO

Rabbit Pvt. Ltd. wants a junior security team to perform a short external reconnaissance and enumeration exercise. Students are not given a target IP in advance. They must first identify a suitable target from Shodan and then perform an authorized, non-destructive assessment of the selected system. The exercise tests networking knowledge, Nmap usage, service enumeration, FTP investigation, prioritization and security reasoning.

Important: Choose only a target that is clearly intended for security research, a CTF/lab system, your own system, or a host for which you have explicit permission to test. Do not scan random public organizations or third-party systems.

OBJECTIVE

Starting from an unknown target, demonstrate the complete workflow: identify a suitable authorized target, understand its network exposure, scan ports, enumerate services, investigate FTP when present, identify a security weakness, and document the reasoning and evidence behind your conclusions.

ASSESSMENT FLOW

Target Selection · Scope · Host Discovery · Port Scanning · Service Detection · Enumeration · FTP Assessment · Security Observation Controlled Validation · Evidence · Report

TASK 1 · TARGET IDENTIFICATION

Use Shodan to identify an appropriate legal training target.

· Search Shodan for a lab/CTF/training host or a system you are authorized to assess.

· Record the target IP, hostname/domain if shown, organization/owner where available, and why you selected it.

· Take a screenshot of the relevant Shodan result.

· Do not treat a Shodan result as proof that testing is authorized.

· Record the exact scope you will test.

Suggested resource: https://www.shodan.io/

Output: Target selection sheet + Shodan screenshot + scope/authorization justification.

TASK 2 · HOST AND PORT DISCOVERY

Perform an initial Nmap assessment against the selected authorized target.

· Run a basic Nmap scan.

· Perform a full TCP scan where appropriate.

· Perform service/version detection.

· Perform OS identification where appropriate.

· Save the scan output.

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· Explain why each scan was chosen.

TASK 3 · BUILD THE ATTACK SURFACE

Convert Nmap results into an attack-surface table. Do not simply paste terminal output.

IP Port Protocol Service Version Priority Reason

Strategy Question: If you discover five services but have limited time, which service will you investigate first and why?

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TASK 4 · SERVICE ENUMERATION

For each important open service, select an appropriate enumeration technique.

· Identify the service and version.

· State what information you need to learn.

· Choose a tool and explain why it is appropriate.

· Run the enumeration.

· Record and interpret the output.

· State the next step based on the result.

Suggested examples: Nmap NSE, curl, WhatWeb, Gobuster, enum4linux, smbclient, FTP client and relevant service tools.

TASK 5 · FTP ASSESSMENT AND CONTROLLED EXPLOITATION

If FTP is exposed, investigate it. If FTP is not present, document that fact and use another discovered service for the same reasoning exercise.

· Identify FTP software/version.

· Check whether anonymous access is allowed.

· Enumerate directories/files where authorized.

· Determine whether download access exists.

· Test upload access only on an intentionally vulnerable lab with instructor permission.

· Use a harmless test file if write access is assessed.

· Explain impact and remediation.

Required flow: Discovery · Enumeration · Weakness · Controlled Validation · Impact · Remediation

TASK 6 · ENUMERATION STRATEGY CHALLENGE

You discover the following services. You cannot fully enumerate all of them within the available time.

Port Service Priority Why? First Enumeration Action

21 FTP

22 SSH

25 SMTP

80 HTTP

445 SMB

161/UDP SNMP

Your strategy must be justified. There is not necessarily one correct priority order.

TASK 7 · UNKNOWN SERVICE INVESTIGATION

During scanning, you discover an unfamiliar service such as 8080/tcp. Determine what is actually running.

· Perform service/version detection.

· Test the service safely using an appropriate protocol-aware client where applicable.

· Use a controlled banner/service check if necessary.

· Identify the technology or application.

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· Explain what you learned and what you would investigate next.

TASK 8 · FINAL MINI ASSESSMENT

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Prepare a concise technical assessment demonstrating the complete chain from target identification to confirmed security observation.

· Target selection and scope

· Host/port discovery

· Service/version identification

· Detailed enumeration

· FTP investigation when available

· At least one prioritized security observation

· Evidence and screenshots

· Impact

· Remediation

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REQUIRED REPORT FORMAT

1. Cover Page
2. Target and Scope
3. Shodan Target Discovery Evidence
4. Methodology
5. Nmap Scanning
6. Attack Surface Table
7. Service Enumeration
8. FTP Assessment / Alternative Service Assessment
9. Strategy Questions
10. Findings and Evidence
11. Impact
12. Remediation
13. Conclusion

EVIDENCE STANDARD

Every important result must contain: Command · Output · Screenshot · What it means · Why it matters · What you did next.

Do not include credentials, personal data or unnecessary sensitive information in the report.

MARKING SCHEME · 50 MARKS

Area	Marks
Target identification, scope and authorization reasoning	5
Network/port scanning and interpretation	10
Service enumeration and attack-surface mapping	10
FTP assessment / alternative service assessment	8
Strategy and prioritization	7
Evidence and technical reasoning	5
Report quality and presentation	5

FINAL INSTRUCTOR NOTE

The assignment evaluates decision-making, not the number of tools used. Students must explain why a scan was chosen, what the output means, what information should be gathered next, and how the finding translates into a security decision.

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